

Big Data Analytics with MongoDB

MongoDB Data Model- Collections

- **Collections** is a set of data organized in a JSON format documents. It is also called Binary Encoding of JSON (BSON) documents

Naming a Collection: A name of collection must obey the following rules:

- It should not contains \$. Or null character
- It should not begins with dot (.)
- It should not be an empty string

Documents: These are data records that organized in the JSON format

```
{
    'student_id' : 1,
    'name' : 'neeraj',
    'age':18, 'mobile': 00000000, 'status': 'active', 'email': 'neeraj@test.t',
    'class_results': [
        { 'class_id': 7 , 'year': 2020, 'marks': 500, 'result': 'pass' }
    ]
}
```

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MongoDB - CRUD Commands

➤ Update Query Examples:

```
db.students.updateMany(  
  {"personal_details.age": {$gte: 18}}  
, { $set: {"adult": "y" } })
```

\$gte: Means greater than or equal

➤ Update QUERIES with \$rename, \$unset Examples:

```
db.students.updateMany(  
  {}, $rename: { "nme": "name" })
```

```
db.students.update(  
  {"personal_details.age": {$gte: 18}}  
, { $unset: {"adult": "" } })
```

```
db.students.update(  
  {"personal_details.age": {$gte: 18}}  
, { $unset: {"adult": "" }}, {mul:  
  true} )
```

MongoDB - CRUD Commands

➤ Update Query with \$min, \$max, \$inc Examples:

```
{ $inc: { <field1>: <amount1>, <field2>: <amount2>, ...  
  } }
```

```
db.students.update( { student_id: 1 }, { $inc: {  
  "class_results.0.marks": 20 } })
```

```
{ $min: { <field1>: <value1>, ... } }
```

```
db.students.update( { student_id: 1 }, { $max: {  
  "class_results.0.marks": 480 } })
```

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MongoDB - CRUD Commands

➤ Delete Queries with Examples:

```
db.students.deleteOne(  
  {student_id: 19 } )
```

```
db.students.deleteMany(  
  {"personal_details.age": {$gte: 18}  
} )
```

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MongoDB - CRUD Commands

➤ Logical Operators Examples:

```
db.students.find( { $and: [{ "name" : "neeraj"},  
{"age" : { $lte : 20 } }] } );
```

```
db.students.find( { $or: [{ "name" : "neeraj"},  
{"age" : { $lte : 20 } }] } );
```

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MongoDB - QUERY Commands

➤ Sort, skip, and limit functions Examples:

```
db.students.find().sort({'name':1})
```

```
db.students.find().sort({'name':1}).limit(2)
```

```
db.students.find().sort({'name':1}).limit(2).skip(1)
```

NOTE: In sorting, “1” stand for ascending order, and “-1” descending order

➤ Nested Fields Examples:

```
db.students.find( { "personal_details.age": 17,  
"personal_details.email": "user7@test.online"} )
```